



Q. All the following are findings of Berger's (IgA) nephropathy except?

- a. Leading cause of glomerulonephritis
- b. Double contour mesangial proliferation
- c. C3 levels are normal
- d. Galactose-deficient IgA1 is hallmark of disease

Ans: (b)



RENAL ARTERY STENOSIS



RENAL ARTERY STENOSIS

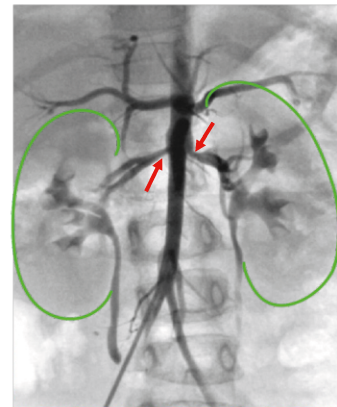
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Classical beaded appearance of Renal artery, seen in FMD (Fibro Muscular Dysplasia)



CT Angiography



Angiography aorta

Ostial narrowing

- Screening test: **Doppler**
- Gold standard investigation: **Renal angiography**
- CT Angiography: Risk of causing contrast induced nephropathy

CAUSES

00:01:51

Mnemonic: FAT

1. **F** - Fibromuscular dysplasia (FMD)

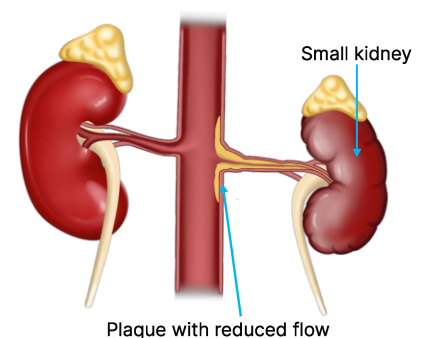
- Rare (not seen in Indian population)
- More common in young females
- Affects distal blood vessels

2. **A** - Atherosclerosis

- M/c cause
- ARAS → 60-year-old male smoker with HTN
- Affected kidney shrinks, contralateral kidney enlarges

3. **T** - Takayasu arteritis

- AKA Pulseless disease
- Seen in Asian population
- Ostial narrowing Renal artery
- Anti-hypertensives need to be used



Important Information

- M/c blood vessel involved in Takayasu arteritis- **Left subclavian artery**

Key word: Flash Pulmonary edema

1. HTN persisting with medication
 - Increasing ACEI dose in undetected B/L RAS leads to AKI
 - ACE inhibitors inhibit efferent arteriole (vasoconstriction) → Loss of pressure gradient (responsible for filtration) in Glomerulus → Development of oliguria/ anuria
2. Abdominal bruit during systole and diastole
 - Intensity ↑ as luminal narrowing ↑
3. Peripheral pulses absent
4. Shortness of breath due to LVF

WORK UP

00:10:10

- Ultrasound
 - Asymmetric kidneys: Size variation **>1.5 cm**
- Doppler
 - Good **screening test**: Obesity reduces the sensitivity
- CT Angiography
 - Small risk of contrast nephropathy
- Magnetic Resonance Angiography (MRA)
 - Expensive
 - Gadolinium induced Nephrogenic systemic fibrosis
- Renal angiography (Invasive test)
 - **Gold standard test**
- Captopril renogram
 - Not done if S. creatinine >2 mg%
- ↑ KFT
- Urine albumin excretion rate
 - Directly related to CV mortality rate

TREATMENT

00:12:11

CONDITION	TREATMENT
Unilateral RAS	• ACEI + CCB + α_1-Blocker
Bilateral RAS	• ACE inhibitors are contraindicated (Trigger AKI) • T/t: PTRAs (Percutaneous Transluminal Renal Angioplasty) + Stenting
Fibromuscular Dysplasia	• PTRAs + Stenting
Atherosclerotic Renal Artery Stenosis (ARAS)	• Medical therapy initially

MALIGNANT HYPERTENSION

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FEATURES

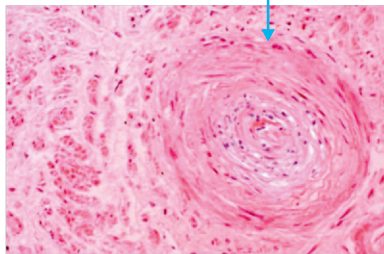
- Disproportionate rise in blood pressure leading to renal malfunction and death
- Mortality rate for 6-12 months is 50%.
- Aka Arteriolonephrosclerosis
- M/c in African-Americans

SYSTEMIC MANIFESTATIONS

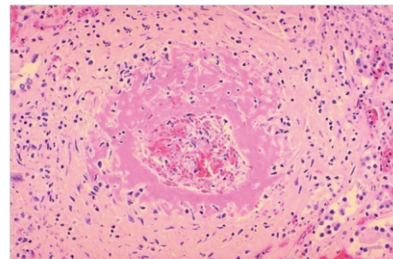
- CNS: Encephalopathy
- Eye: Fundus → Retinal hemorrhage
- ↑ KFT
- Microangiopathic hemolytic anemia (Shear stress on RBCs): **Schistocytes** on peripheral smear

HISTOPATHOLOGY FINDINGS

Onion skin appearance



Hyperplastic arteriosclerosis



Fibrinoid necrosis of small renal arteries

MANAGEMENT

- Aggressive BP control

ATHEROEMBOLIC KIDNEY DISEASE

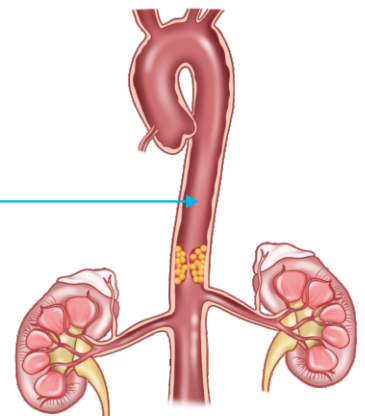
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- Sudden derangement of KFT post angiography (immediately or 1-14 days after procedure)

MANIFESTATIONS

- AKI- Anuria
- Toe gangrene
- Fever, abdominal pain
- Urine microscopy findings : **Eosinophiluria**

Passing the guide wire through the plaque causes fragments to enter RA



POSTMORTEM KIDNEY BIOPSY FINDING

- Microvessel occlusion by cholesterol crystals

Important Information

- PAN (Polyarteritis nodosa): Not a cause of Renal artery stenosis
 - Formation of microaneurysms



Q. A 20-year-old female patient presents to the clinic with complaints of fatigue, muscle weakness, and occasional palpitations. She has a history of poorly controlled hypertension despite being on medications. An ultrasound examination of the kidneys reveals asymmetric kidneys, with one kidney appearing smaller in size while the contralateral kidney appears enlarged. Which of the following is the likely cause contributing to the patient's symptoms?

- a. Atherosclerosis
- b. Fibromuscular dysplasia (FMD)
- c. Takayasu arteritis
- d. Polyarteritis nodosa

Ans. (b)

Q. A 55-year-old Afro-American male presents to the emergency department with severe headaches, confusion, and visual disturbances. On examination, his blood pressure is significantly elevated. Fundus examination reveals retinal hemorrhage, and laboratory investigations show evidence of microangiopathic hemolytic anemia on peripheral smear. Based on these findings, which of the following conditions is most likely responsible for the patient's presentation?

- a. Diabetic nephropathy
- b. Chronic glomerulonephritis
- c. Malignant hypertension
- d. Polycystic kidney disease

Ans. (c)

- In a lower GI bleed with hemodynamic compromise, Initial investigation: **Upper GI endoscopy**

CLASSIFICATION AND SCORES

00:01:28

Glasgow Blatchford Classification	- Predicts the need for hospitalization - Value > 2 → Require hospitalization
AIMS65 Score	- Predicts mortality - Indicates need for endoscopic therapy
Forrest Classification	Decides the type of endoscopic therapy needed in peptic ulcer bleeding based on endoscopic findings

INTRODUCTION

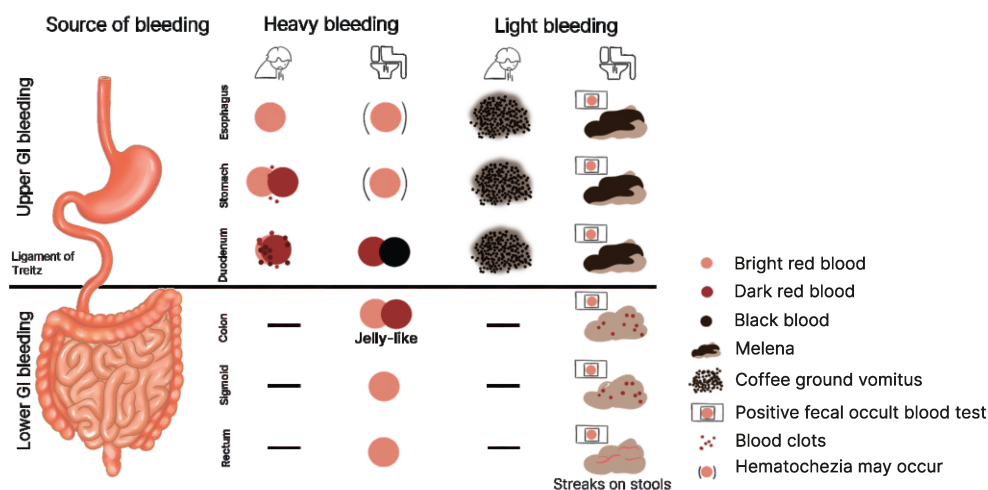
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- Anatomical landmark for upper GI & Lower GI bleeding - **Ligament of Treitz**
 - Proximal to ligament of Treitz → Upper GI bleeding
 - Distal to ligament of Treitz → Lower GI bleeding
- Location of ligament of Treitz → **Duodenojejunal flexure**
- Area distal to ligament of Treitz - Scanned by Verlac Capsule Endoscopy or MR enterography or paediatric colonoscope or deep enteroscopy

PRESENTATION OF UPPER GI BLEEDING

00:06:02

- Hematemesis:
 - Initial episode: Bright red vomitus
 - 2nd or 3rd episode - **Coffee grounds vomitus** (Brownish vomitus)
 - E.g., In portal HTN oesophageal varices may rupture → some blood due to gravity goes into stomach where acid degrades Hb to produce acid hematin (brownish in colour)
- Haematochezia with hypotension & drop of Hb in > 48-72 hours
 - Do Upper GI endoscopy



• For a need for hospitalization in upper GI bleeding

SCORE	DECISION
0-1	No need to admit
2	Admit
>7	Urgent Endoscopic therapy & ICU care

Admission risk marker	Score
Blood urea (mg/dL)	
• 18.2~22.3	2
• 22.4~27.9	3
• 28.0~69.9	4
• ≥70	6
Hemoglobin (g/dL)	
• Men	
○ 12~13	1
○ 10~11.3	3
○ <10	6
• Women	
○ 10~12	1
○ <10	6
Systolic blood pressure (mmHg)	
• 100~109	1
• 90~99	2
• <90	3
Pulse ≥100/min	1
History and comorbidities	
• Melena	1
• Syncope	2
• Hepatic disease	2
• Cardiac failure	2

INITIAL INTERVENTIONS IN PATIENT WITH HEMATEMESIS

00:09:15

- Fluid resuscitation: Target blood pressure < 90/60 mmHg
 - Blood grouping & cross-matching should be done
 - Blood transfusion is not done without fixing the source of bleeding
- PPI infusion: For peptic ulcer
 - After 2-3 days, step down to a potassium competitive acid blocker, i.e., Vonoprazan or to oral PPI's
- Proton pump inhibitor (PPI) drug: IV erythromycin to improve visualization, 30-60 min before performing upper GI endoscopy
- Alcoholic cirrhosis:
 - With ascites → ↑Risk of spontaneous bacterial endocarditis → **Ceftriaxone**
 - Portal HTN → Octreotide (infusion), terlipressin (bolus)
- M/c cause of hematemesis – **Peptic ulcer disease**
- Upper GI Endoscopic findings that indicate aggressive intervention
 - Active bleeding/visible bleeding vessel
 - Adherent clot

INTERVENTIONS

- Endoscopic therapy
 - Inject Adrenaline (local administration) → vasoconstriction → reduce amount of bleeding
 - Hemostatic clips → fix the bleeding blood vessel
 - Thermal coagulation: limited coagulation d/t thin duodenal wall
 - Can cause extensive damage & perforation of bowel
- PPI infusion – To increase intragastric pH to ≥ 6.0
- Diet post-endoscopic procedure: Clear fluids
- Discharge the patient by 3rd or 4th day

FOREST GRADING

00:15:37

- Done to decide whether there's a need for endoscopic therapy

FOREST GRADE	ENDOSCOPIC FINDING	REBLEEDING RISK	ENDOSCOPIC THERAPY IF NEEDED
IA	Spurting arterial blood	90%	Yes (Urgent)
IB	Oozing hemorrhage	30-50%	Yes
IIA	Visible vessel (non-bleeding)	40-50%	Yes
IIB	Adherent clot	20-30%	Yes (consider clot removal first)
IIC	Flat pigmented spot	10%	No (medical therapy done)
III	Clean-based ulcer	5%	No (PPI therapy only)

AIMS625 SCORE

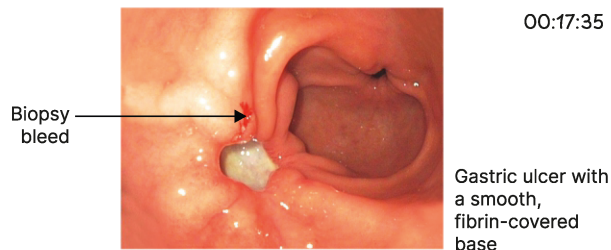
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- Predict chances of mortality

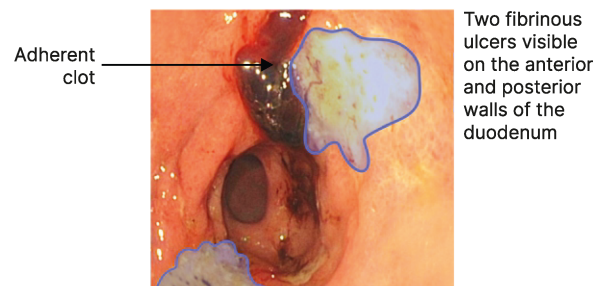
Albumin	< 3g/dl
INR	>1.5
Mental Status (checked d/t cerebral hypoperfusion)	GCS < 14
SBP	<90mmHg
Age	> 65

PEPTIC ULCER DISEASE

- Source of bleeding in gastric ulcer: **Left gastric artery**
 - M/c site of gastric ulcer – **Lesser curvature**
- Source of bleeding in duodenal ulcer: **Gastrooduodenal artery**
 - M/c site of duodenal ulcer – 1st part of duodenum i.e., duodenal cap
- Anterior duodenal ulcer – Most likely to perforate
- Posterior duodenal ulcer – Most likely to bleed



Gastric Ulcer



Duodenal Ulcer

EROSIVE GASTRITIS

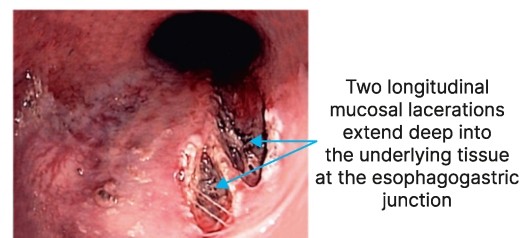
- ↓ Prostaglandins: Due to
 - Long-term intake of COX 10
 - Trauma, burns (curling ulcer), raised ICP (cushing ulcer), patient on ventilator or ARDS
 - T/t: Pantoprazole

ESOPHAGEAL VARICES

- Portal HTN – H.V.P.G (Hepato-venous pressure gradient) > 5 mmHg.
- Bleeding risk, when H.V.P.G > 12 mmHg
- **Source of bleeding** – **Left gastric vein** (branch of portal vein) & azygous vein
 - D/t development of Porto-systemic collaterals
- Leading cause of portal HTN in adults → **Alcoholic or non-alcoholic cirrhosis**
- Leading cause of portal HTN in children → **EHPVO** (Extra Hepatic Portal Vein Obstruction)
 - Umbilical vein catheterization in a neonate → trauma → thrombus formation → EHPVO
- **TOC** for bleeding esophageal varices: Endoscopic Variceal Ligation + Vasoactive medication
- Prevention of bleeding from esophageal varices – By Endoscopic Variceal Ligation > Nadolol (↓ pressure in portal vein)

MALLORY-WEISS SYNDROME

- **Submucosal tear** (Non-life-threatening condition)
- Associated with:
 - Alcohol: binge drinking → Retching → Hematemesis
 - Hyperemesis gravidarum → Retching → Hematemesis
- Site: **Gastric side of gastroesophageal junction**
- Source: Left gastric artery
- Treatment:
 - Active bleeding: Endoscopic therapy (Hemostatic clips), antiemetics, clear liquids × 1 day
 - No active bleed: No endoscopic therapy, antiemetics, & regular diet



Mallory-Weiss syndrome



EXTRA MILE

Free By - ZeroTrace

MACKLER'S TRIAD

Forceful vomiting

Chest pain: Bursting of esophagus → Chemical mediastinitis

Subcutaneous emphysema:
Due to escape of air from esophagus into the mediastinum

- **Stomach volvulus**

- Stomach fold on its own axis

BORCHARDT'S TRIAD

Violent retching but no vomiting

Acute epigastric pain

Inability to pass NG tube

DIEULAFOY'S LESION

- Enlarged submucosal artery → **Punctate erosion** in otherwise normal stomach mucosa
- Site: <6 cm of lower esophageal junction on the lesser curvature
- Vascular malformation → Can be seen in the rectum, colon, and small bowel
- **T/t: Endoscopic hemostatic clips**

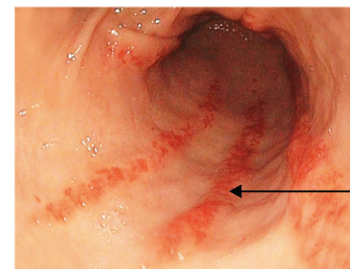
Important Information

- **Heyde Syndrome:** Aortic stenosis + Bleeding vascular ectasia (colonic) + Acquired vWD
 - Aortic stenosis – causes shear stress on RBCs & multimers on vWF.
 - Fragmentation of long chain of vWF → Deficiency of vWF (Acquired vWD)

SUBSEQUENT CAUSES

00:30:57

- Neoplasm: GIST, lymphoma, adenocarcinoma stomach
 - GIST
 - If localized → surgical resection
 - Origin → Interstitial cells of Cajal (Stromal tumour)
 - Metastasis to liver → Tyrosine kinase inhibitors like imatinib
- GAVE: Gastric Antral Vascular Ectasia
 - Upper GI endoscopy finding → **Watermelon stomach**
- Prolapse gastropathy: Stomach prolapses into esophagus

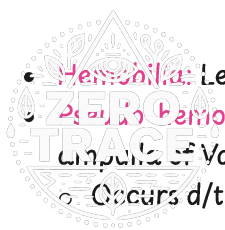


GAVE: Watermelon Stomach

Fragile dilated blood vessels are visible as red stripes in the pyloric antrum

Important Information

- Portal gastropathy: Seen in patients with portal HTN
 - Bleeding from esophageal varices & congested mucosa of stomach



- **Hemobilia:** Leading cause – Iatrogenic Trauma (Laparoscopic cholecystectomy)
- **Hemobilia/Hemosuccus pancreaticus:** Bleeding from pancreatic duct into 2nd part of duodenum via **ampulla of Vater**
 - Occurs d/t rupture of pseudoaneurysm of peripancreatic vessel

SMALL INTESTINAL SOURCES OF GI BLEEDING

00:35:26

- Vascular ectasia, Dieulafoy's lesion
- Malignancy: Adenocarcinoma
- NSAIDs: COX 1 Inhibitors
- Meckel's diverticulum
 - Scan used - **Tc 99m pertechnetate**
 - Rule of 2
 - 2 inches in size
 - 2 feet from ileocecal junction
 - 2 mucosae (stomach mucosa, pancreatic mucosa)
- Crohn's disease: Deep serpiginous ulcers

WORK UP

00:37:50

- CTA, angiography → Detect vascular malformation
 - Tc99 labelled RBC scan: Detect vascular malformation & Mickels diverticulum
 - WCE (wireless capsule endoscopy)
 - MRE/ CTE
 - Deep enteroscopy
- } For Crohn's disease

LOWER GI BLEEDING

00:38:20

CAUSES

NOT REQUIRING HOSPITALIZATION

Internal haemorrhoids	Anal fissure
• Source: Superior rectal vein • IOC: Proctoscopy • In primary piles – Veins are seen at 3'o clock, 7'o clock & 11'o clock position • No spasm of anal sphincter • Flash in pan: Spurting of blood after defecation	• M/c location: Midline posterior • T/t: Topical nitro-glycerine/ Calcium channel blockers/ lignocaine • Spasm of anal sphincter • Streaking of stool with blood

REQUIRING HOSPITALIZATION

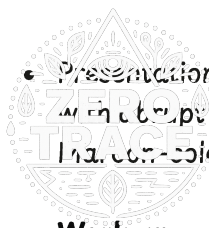
DIVERTICULOSIS

- Areas of weakness, where vasa recta penetrate bowel
- M/c site: Left side (Sigmoid colon/left side of colon)
- Right side diverticulosis: ↑ incidence of bleeding
- Source: Branches of the marginal artery (Vasa recta)
- Ba enema: **Saw-tooth appearance**

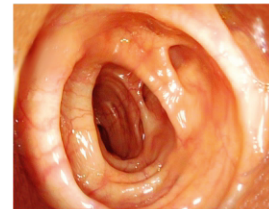


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Ba enema: Saw-tooth appearance



- **Presentation:** Geriatric patient, American/European person presents with abrupt onset of painless hematochezia /
Malcoloured stool → dizziness or syncopal episodes (severe blood loss)



Several small, circular, blind outpouchings of the mucosa

Diverticulosis

- **Work up**

- Hemodynamically stable patient + prep patient with polyethylene glycol: Colonoscopy
- Hemodynamic unstable patient: CTA, Angiography
- 90% cases: Spontaneous resolution
- Ongoing bleed requires **angiographic embolization** or endoscopic therapy

Important Information

- Saw tooth appearance in Ba enema: Diverticulosis
- Saw tooth EEG: REM sleep
- Saw tooth ECG: Atrial flutter

INFLAMMATORY BOWEL DISEASE: UC/CD

- Montreal classification: Classification based on extent & severity of disease
 - Inflammation involving on rectum: E1
 - Inflammation up to splenic flexure: E2
 - Inflammation beyond splenic flexure: E3
- Test done for
 - Crohn's disease: Wireless capsule endoscopy > MR enterography
 - Ulcerative colitis: Colonoscopy/sigmoidoscopy

MALIGNANCY

- **Ca caecum:** Higher chances of bleeding
 - Stool occult blood test → positive
 - Progressive anaemia
- **Ca rectum**
 - Right side colon cancer mainly presents → bleeding
 - Left side colon cancer mainly presents → obstructive symptoms

ISCHEMIC COLITIS

- Risk factors: Illicit drugs like amphetamines, endurance sports, chronic constipation
- X-ray abdomen: **Thumbprint pattern**
- **IOC:** Colonoscopy

OTHER CAUSES

- NSAID-induced colitis
- Rectal varices
- Vasculitis, Aortic-colic fistula
- **Angiodysplasia of colon:** Vascular malformation
 - Colonoscopy: Normal (multiple times)
 - CTA, Angiography

00:47:37



- **1st Vascular cause of Lower GI Bleed**
- **Hughes Syndrome:** Aortic stenosis (↓ VWF) + Colonic Vascular ectasia
- **Systemic causes:** Uremia & liver failure

WORK UP LOWER GI BLEEDING

00:48:36

SYSTOLIC BP ↓↓	SBP NORMAL
• 1st investigation: Upper GI endoscopy • If upper GI endoscopy is negative: ○ Able to prep with PEG: Colonoscopy ○ Unable to prep: CTA, Angiography,	• <40 yr: Flexible sigmoidoscopy • >40 yr: Colonoscopy

MCQ's

- Q. A 45-year-old alcoholic male presents to the emergency department with severe abdominal pain and feeling lightheaded. Examination shows a non-cooperative patient with rebound tenderness over epigastrium. Bowel sounds are reduced. Lab shows haemoglobin of 10 gm/dl with increased TLC and deranged KFT. Which is likely cause of this presentation?
- Viscous bleeding
 - Mallory-Weiss Syndrome
 - Bowel Perforation
 - Portal Hypertension

Ans. (c)



PEPTIC ULCER DISEASE

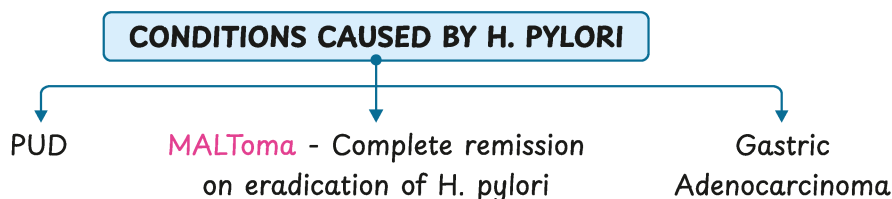


PEPTIC ULCER DISEASE

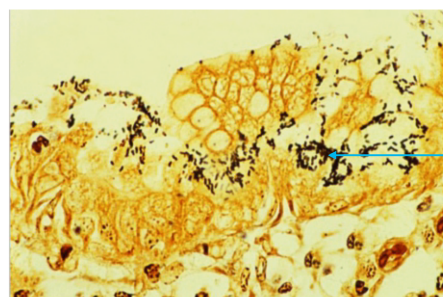
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- M/c cause of PUD: *H. pylori* (Type B Gastritis)

H. PYLORI

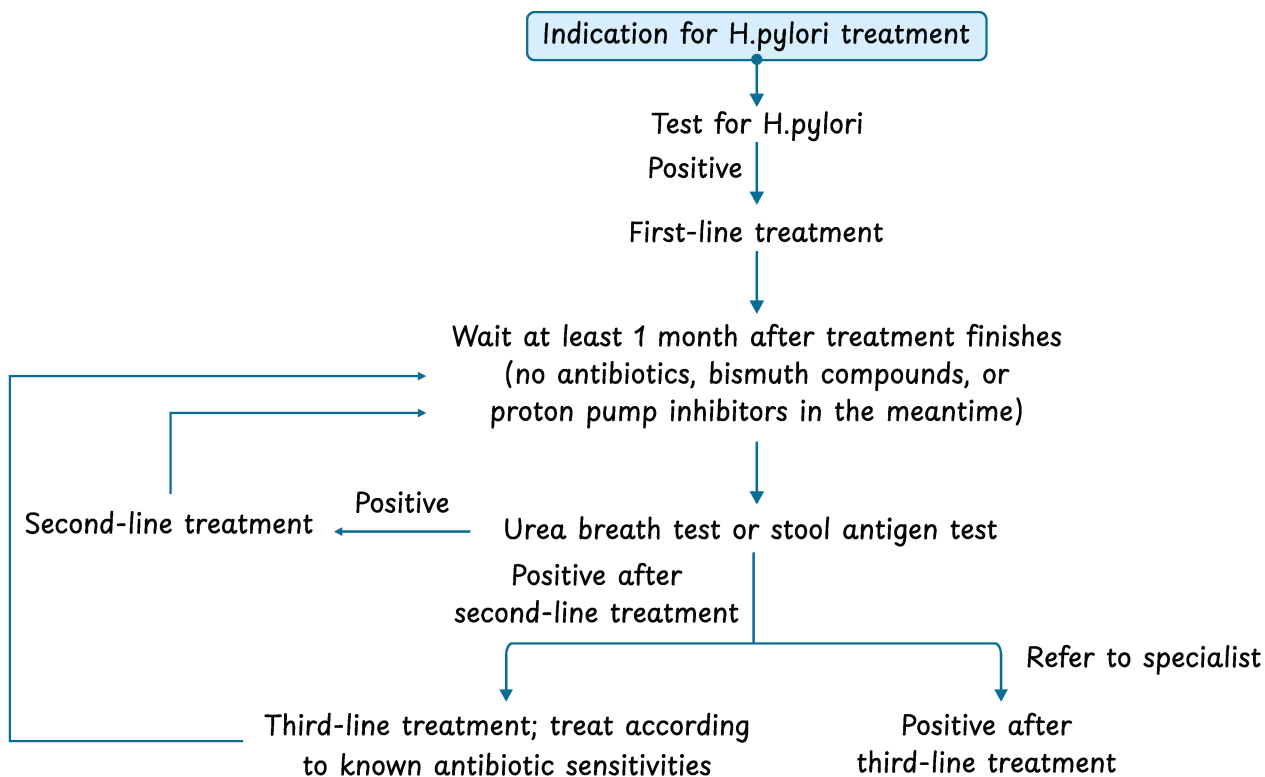


- Tests for detection of *H. pylori*:
 - Earlier:
 - Stomach biopsy, stained with *Warthin starry stain*
 - Currently: non-invasive tests
 - Urea Breath Test (UBT) - better than stool antigen test
 - Stool Antigen Test (Validated)



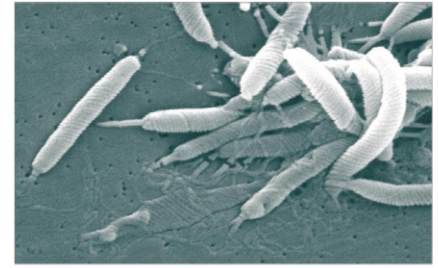
H. Pylori

Warthin starry stain



FEATURES OF H. PYLORI

- Urease-producing, gram-negative, spiral-shaped bacteria
- Non-spore-forming (eradication is possible), can transform into coccoid shape
- Spread - feco-oral
- Extracellular bacteria - Colonizes antrum and corpus
- **Microaerophilic**
- Urease-producing bacteria



Metabolizes urea → Ammonia + Co₂

- Alkaline pH of ammonia causes
 - Stimulation of Gastric cells → **Gastrin** ↑
 - Stimulation of Parietal cells → **Acid** ↑ (**Hyperchlorhydria**) → predisposes to Duodenal ulcer > gastric ulcer

VIRULENCE FACTORS OF H. PYLORI

VIRULENCE FACTOR	FUNCTION
Urease	Converts urea → ammonia → neutralizes gastric acid around bacterium
Flagella	Helps motility through thick gastric mucus
Cag A (Cytotoxin-associated gene A)	Associated with inflammation and increased cancer risk
Vac A (Vacuolating cytotoxin A)	Damages gastric epithelial cells

WORK-UP FOR H. PYLORI

00:09:58

NON-INVASIVE TESTS	INVASIVE TESTS
• Urea Breath Test (UBT): Best • Stool Antigen Test ○ Perform stool antigen test 4 weeks after stopping antibiotics ○ Stop antisecretory agents ~7 days before UBT	• Urease Test (CLO Test) • Stomach Biopsy

- Both UBT and Urease can be false negative d/t recent use of:
 - PPIs, Antibiotics, Bismuth
- Best for **detection/eradication** of H. Pylori: UBT
- **IOC** to document PUD: Upper GI endoscopy

T/t

TRIPLE THERAPY	• Clarithromycin(2 weeks) + Amoxicillin (2 weeks) + PPI (6 weeks)
QUADRUPLE THERAPY	• PPI + Bismuth + Tetracycline + Metronidazole ○ Bismuth: surface protective agent → ↓ acid-mucosa contact time



HYBRID THERAPY

- PPI + Amoxicillin × 7 days F/b
- PPI + Amoxicillin + Clarithromycin + Nitazoxanide

LOAD

- Levofloxacin + Omeprazole + Alinia (Nitazoxanide) + Doxycycline

Important Information

- Erythromycin is not effective for eradication of *H. pylori*

TYPE A GASTRITIS

00:16:32

- Autoimmune
- Antibodies - Anti-intrinsic factor and anti-parietal cell
- Intrinsic factor ↓, B12 ↓, Pernicious anemia
- Achlorhydria
- Autoimmunity damages stomach mucosa leading to gastric ulcer

	GU LOCATION	GASTRIC ACID LEVEL
TYPE 1 (M/c)	Lesser curvature/Gastric body	Low
TYPE 2	Antrum	Low/Normal
TYPE 3	<3 cm of pylorus (associated with DU)	High/Normal
TYPE 4	Cardia	Low

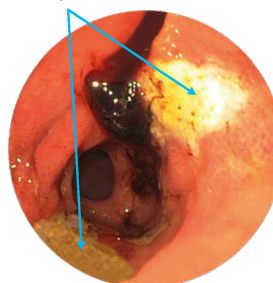
GASTRIC ULCER VS DUODENAL ULCER

GASTRIC ULCER (GU)	DUODENAL ULCER (DU)
• Blood Group A • > 6th decade • Premalignant condition • Lesser curvature	• Blood Group O • Basal/Nocturnal acid production ↑ • D1 or duodenal cap • Size of ulcer: <1cm • Giant DU - size >2cm

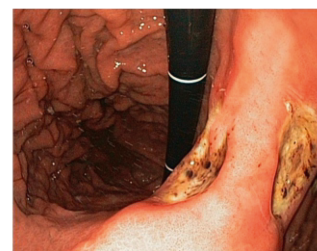
COMMON FEATURES

- Epigastric pain (gnawing sensation) & tenderness
- Tachycardia/orthostatic hypotension - Vomiting, GI bleed
- Rigidity of anterior abdominal wall - Perforation of viscus
- **Succussion splash** - Gastric outlet obstruction

Kissing lesions



Duodenal Ulcer: Endoscopic view



Gastric Ulcer: Endoscopic view

Life Complication of PUD

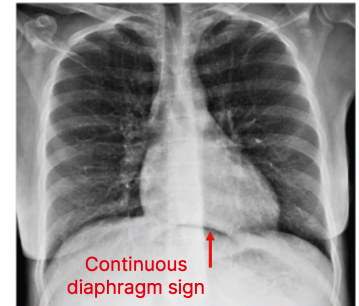
- Gastric Ulcer: Source of bleed → **Left gastric artery**
- Duodenal Ulcer: Source of bleed → **Gastroduodenal artery** (Posterior > Anterior)
- ↑ Risk in Elderly patients and concomitant NSAID use
- 30-day mortality: **2.5-10%**
- **FORREST** Classification : Used to stratify bleeding risk and guide interventions
- Leading cause of death: Non-bleeding causes
 - E.g. - Multi organ dysfunction, pulmonary complications

PERFORATION

- 30-day mortality - **>20%**
- **Triad**: Acute abdominal pain + HR ↑ + Rigidity of anterior abdominal wall
- On X-ray: gas under diaphragm
 - **Cupola sign**
 - **Moustache Sign**



X-ray shown gas under diaphragm



X-ray showing pneumomediastinum

PENETRATION

- Penetration - slow gradual erosion
 - DU: erode into pancreas
 - GU: erode into left lobe of liver

GASTRIC OUTLET OBSTRUCTION

- Ulcer related inflammation and local edema in prepyloric area d/t duodenal ulcers
- Late stages - Scar formation
- There will be Chronic vomiting leading to **Hypochloremic Hypokalemic metabolic alkalosis**

GASTROCOLIC FISTULA

- Mortality: >20%

SIDE EFFECTS OF PPIs

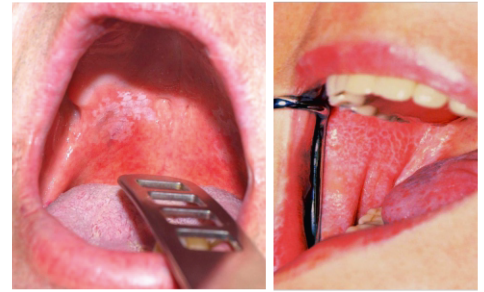
Mnemonic : **ABCD-M**

- **A**IN: Eosinophilia and eosinophiluria
 - Triad: fever + rash + AKI
 - Urine M/E - Eosinophiluria
- **B**one pain
- **C**lostridium **D**ifficile ↑ infection
- **C**AP ↑ incidence
- CKD (PPI can act as hapten, recognised by APC dendritic cells as foreign)
- Mg ↓ (d/t impaired gut absorption)

PREMALIGNANT CONDITIONS OF GUT

00:30:52

- **Oral Submucous Fibrosis:** d/t betel nut consumption
- Leukoplakia, Erythroplakia
- Syphilitic glossitis
- Oral lichen planus (**least** chances) - Wickham striae (criss-cross striations)

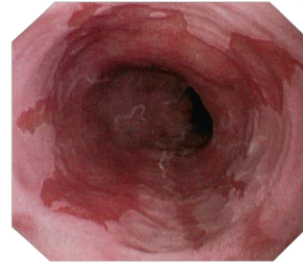


Leukoplakia

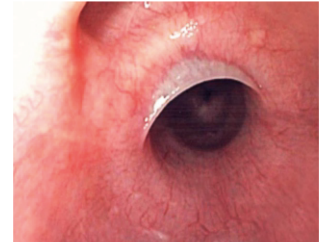
Lichen Planus

ESOPHAGUS

- Tylosis Palmaris - Hyperkeratosis of palms/soles, ↑ risk of squamous cell carcinoma
- Achalasia Cardia, Barrett Esophagus
- Post cricoid dysphagia/Plummer Vinson syndrome
 - Esophageal web (upper esophagus)
 - Iron-deficiency anemia (IDA)
 - Koilonychia
 - ±Glossitis



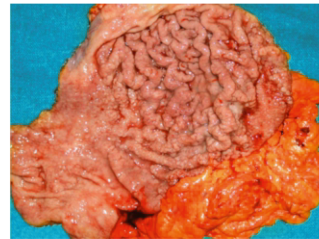
Barrett Esophagus



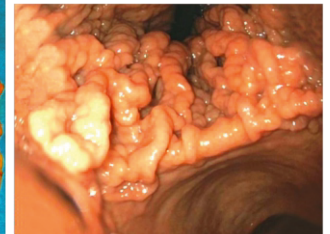
Esophageal web

STOMACH

- Type A Gastritis, Gastric Ulcer
- Ménétrier's Disease
 - Foveolar cell hyperplasia + Parietal cell atrophy
 - Gastric ulcer
 - Protein-losing enteropathy:
 - Diarrhea + pedal edema (d/t hypoalbuminemia)
 - UGIE: "**Cerebriform**" appearance of stomach mucosa
 - T/t - **Cetuximab**



Cerebriform appearance



Menetrier's disease

SMALL INTESTINE

- Crohn's disease

LARGE INTESTINE

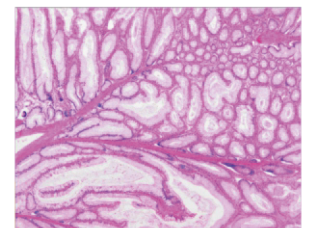
- IBD - UC
- FAP: **APC gene, chromosome 5**, >100 polyps/colon
 - 1° - Prophylactic Colectomy

PEUTZ-JEGHERS SYNDROME

- **AD, STK11/LKB1** gene on chromosome **19** (serine threonine kinase)
- Mucosal melanosis on Lips
- Hamartomatous polyps - jejunum
- ↑ Risk of cancer - Ovary, Cervix, Hepatobiliary
- HPE: **Arborising pattern**, smooth muscle hyperplasia

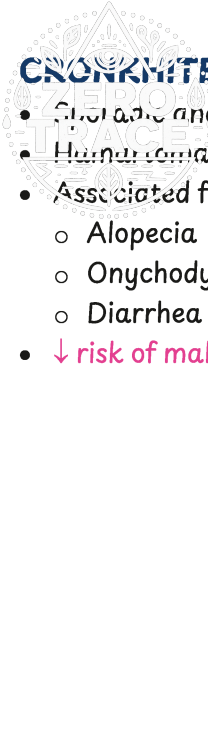


Mucosal Melanosis



Hamartomatous Polyp

00:38:16



CROHNITE-CANADA SYNDROME

00:39:45

- Gradual and more seen in Japanese population
- Hamman-Rich polyps - Entire gut except for esophagus
- Associated findings:
 - Alopecia
 - Onychodystrophy
 - Diarrhea
- ↓ risk of malignant transformation



Q. A 35-year-old patient has been diagnosed with Peutz-Jeghers Syndrome. What is the most common site for the development of polyps seen in this condition?

- a. Colon
- b. Duodenum
- c. Jejunum
- d. Ileum

Ans. (c)

Q. Which artery is the most common source of bleeding in duodenal ulcer?

- a. Left gastric artery
- b. Superior mesenteric artery
- c. Gastroduodenal artery
- d. Splenic artery

Ans. (c)